

Chapter 3

Organization and Strategy for University-Industry Relationships



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3.1 Introduction

Universities have long held three missions: service to society, knowledge creation, and education. Toward the first aim, American higher education embraced economic development, especially since the latter part of the twentieth century. Governmental support to establish land-grant universities required new forms of practical training, as well as research and development (R&D) that catalyzed modernization in agriculture and creation of new manufacturing industries. The Hatch Act of 1887 took a step further through agricultural experiment stations directly serving farms and food processors. As universities grew their research programs, particularly in the post World War II era, society has increasingly looked to them as a source for technological innovations that improve people's lives through health, security, and general well-being. Universities have responded through investments in laboratories, faculty, and students along with developing a support structure to facilitate University-Industry (UI) engagement. This chapter focuses on the ways that universities have organized their UI engagement programs, along with their scope, values, and authority.

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3.2 Challenges and Opportunities

UI engagement operates within the frameworks established by governments, influenced by financial incentives as well as regulation. In the twentieth century, the United States' Bayh-Dole Act conveyed ownership of intellectual property (IP) derived from federally funded research to the colleges and universities that conduct the research, and also established regulations as to how IP is commercialized. Various federal programs, such as the National Science Foundation's Engineering Research Centers program, incentivized industry/university collaborations in research. These synergistic activities are reflected in the "triple helix" of university-industry-government collaboration, as described by Etzkowitz and Leydesdorff (1995). Universities conduct fundamental research with government support and subsequently commercialize that research through technology transfer with companies, both established corporations and start-ups (Cole, 2009; Thorp & Goldstein, 2013; Geiger & Sa, 2008).

Whereas technology commercialization is not new for universities, American higher education is increasingly making industry engagement and economic development central to their strategy. The language of "innovation" and "entrepreneurship" has been elevated in strategic plans (Hall & Lulich, 2021) over the last five years, as universities have increasingly recognized that spinning off companies, commercializing technology, and improving economies were critical to their stakeholders. Simultaneously, university associations, such as the Association of Public and Land-grant Universities (APLU), have prioritized contributions to economies as member goals (APLU, 2014; APLU and UEDA, 2015). This trend is not unique to the United States, as indicated in Maassen et al. (2019), which found a global trend toward a "third mission," entailing "mutually beneficial partnerships to social and cultural needs in society, to demands from politics and the economy."

High impact UI engagement requires planning and perseverance as it demands adaptation of university culture and creation of new offices and organizational structures. In part, adaptation has been a matter of creating a new mindset, allowing and incentivizing individuals and academic units to pursue nontraditional industry funding, as in the "entrepreneurial university" model of Etzkowitz et al. (2000). Adaptation has also meant establishing offices that seek out and support industry collaborations in new ways. Simultaneously, universities struggle with how to proactively support industry while maintaining objectivity. This has required careful review of policies that define acceptable behavior (e.g., conflicts of interest), rights (e.g., academic freedom), and rewards (e.g., tenure and promotion standards).

3.3 Prior Research

The literature on UI engagement is extensive, as surveyed by Ankrah and Al-Tabbaa (2015), Geuna and Muscio (2009), Orazbayeva et al. (2019), Perkmann et al. (2011, 2013, 2021), Rothaermel et al. (2007), Rybníček and Königsguber (2019), and Schofield (2013). Taken as a whole, prior research has examined practices and organizations for knowledge transfer and sharing among academic researchers and industry practitioners, through the lenses of incentives, outcomes, and barriers. As shown in these surveys, numerous studies have investigated factors that lead to collaborative success (or failure), as reflected in quantified Key-Performance-Indicators (KPIs), at the level of projects, organizations, and multi-partner alliances.

As noted by Perkmann et al. (2013), “engagement is less organizationally embedded than commercialization, and is more autonomously driven by individuals.” Whereas the process of technology transfer through intellectual property protection and licensing typically is managed through a central university office, engagement is multifaceted, taking many transactional forms, sometimes centralized and sometimes localized within schools, centers, or academic departments. The multidimensionality of engagement is reflected in Schofield’s review (2013), which applies a framework developed by Pertuze et al. (2010), examining institutional and individual success factors, including technical, process, and financial relationships. Furthering this direction, Ankrah and Al-Tabbaa (2015) documented the great range of organization forms, relationships, and degree of formalization in UI engagement.

KPIs are a byproduct of goals and purpose, which have been studied from perspectives of government policy, institutions and individuals. In the United States, APLU’s Economic Development and Community Engagement program reflects the priority of America’s public institutions to support job creation and economic advancement. The program conveys a designation of “Innovation and Economic Prosperity University” from peer-reviewed self-assessments that document growth and improvement in economic engagement, “making the products of learning, discovery, and engagement relevant and useful” (APLU, 2021, p. 16). APLU’s program reflects, in part, expectations of the state governments that fund public universities. Outside of the United States, Dollinger et al. (2018) provide an international comparison of UI goals among seven countries within entrepreneurial, technical, and managerial components, each in the context of government policy.

At the individual level, D’este and Perkmann’s U.K. study (2010) of participants in UI collaborations found that academics were motivated to further their research more so than commercialization. Ankrah et al. (2013) developed a set of case studies surrounding Farraday Partnerships, examining motives, beneficial outcomes and drawbacks of participants from both universities and industry partners. The authors found that motivations differed between industry and university actors, but typically did not conflict with each other. Example motivations included access to state-of-the-art techniques or equipment, joint publications, training opportunities for students (organizational benefits); economic development/technology transfer (societal benefits); and government support or commercialization (financial benefits).

Industry engagement is not strictly limited to knowledge and technology creation and sharing, but may also extend to engagement through work in the forms of internships, cooperative education, and industry relevant courses (Ferns et al., 2016, for example).

Bruneel et al. (2010) found that UI engagement demands effort to overcome two types of barriers: differences in orientation and conflicts in transactions. As the authors state, “UI collaborations are likely to be plagued with conflicts due to a weak attitudinal alignment between partners,” particularly with respect to publication of research findings. They also highlight conflicts over intellectual property terms as a transactional barrier. Prigge (2005) offers strategies to guide universities to mitigate perceived risk that industry relationships may harm academic standards.

Jonsson et al. (2015) examine employment of “professional innovation support officers” as intermediaries that bridge the academic world and companies, providing the bridge to build trust among participants (see also Gulán-Muros et al. (2017), Wright et al. (2008), and Yusuf (2008) for research on intermediaries). Acworth (2008) describes the Cambridge-MIT Institute’s “Knowledge Integration Community” (KIC) model, which coalesces UI engagement around grand challenges in science or technology. The KIC model is a multidimensional alliance integrating education, research, industry, and government. Within each KIC, a manager is responsible for furthering knowledge exchange and, toward that end, is given status equal to the faculty principal investigators from Cambridge and MIT. At a more operational level, research by Debackere and Veugelers (2005) examined how technology transfer offices can serve as “information science links” between decentralized research groups and industry partners, blending the roles of an intermediary and transaction specialist focused on licensing.

Organizationally, UI engagement takes many forms. In one dimension, knowledge transfer offices (KTOs, often called technology transfer offices and other names) offer an interface between the university research enterprise and private companies. Brescia et al. (2016) investigated organization of knowledge transfer among 200 highly ranked universities, evaluating contextual factors that may explain adoption of internal, finding significant variety in approaches.

Industry and universities interact in many ways, both formalized through agreements and informal, through literature, conferences, conversations with former students, for example. Formalized relationships may be institutional (such as sponsored research, technology licenses, joint ventures, and so on) or personal (consulting or start-up investment). Cohen et al. (2002) document these various channels.

Pertuze et al. (2010) recommend as best practices that: UI relationships build from a strategic context; include boundary spanning project managers as intermediaries; communicate the company vision to the university research team; establish strong communication; build broad awareness in the company; and support the work until the research can be exploited. Ulrichsen (2019) offers steps to strengthen UI partnerships through regional consortia that build capabilities and talent, linking education and research within industry clusters, pursued through many avenues and organizational structures. As noted by Frolund and Riedel (2018), collaborations may entail hackathons, innovation boot camps, idea contests, exposure to

technology trends, research-based start-ups in addition to sponsored research projects. Prigge and Torraco also find that successful partnerships come in many forms, including trained labor, classroom experiences, and identification of “significant, interesting and relevant” problems. Given the large set of options available for UI relationships, Awasthy et al. (2020) propose a flexible approach through which UI interactions are matched to the motivations and capabilities of the stakeholders, and a legal framework for cooperation. The breadth and continuum of UI engagement strategies is further reflected in the Partnership Continuum guide UIDP (2012), which addresses the multifaceted nature of engagement strategies.

In summary, UI engagement takes many forms with diverse goals. Knowledge sharing and technology transfer are common components, but the exact form varies over a spectrum from informal conversations to highly structured contractual agreements. The vector for knowledge sharing may be open and available to all, such as published literature, conference presentations, or publically shared resources, such as data; the vector may be through human interaction and collaboration, such as students, alumni, faculty or industry practitioners, which are sometimes constrained by nondisclosure-agreements. On the other hand, limited research exists on the totality of UI engagement with industry in all of their dimensions. Galán-Muros et al. (2017) is one example, representing three domains of interaction (education, research, and “valorization”) along with associated activities. In the following sections, we go further by developing the dimensions for UI engagement, and analyzing how American universities have been organized to address these dimensions.

3.4 Approach

UI engagement was evaluated in three steps, focused on members of UIDP which offers the opportunity to engage individuals who are highly experienced in multifaceted relationships. They also represent prominent institutions, both at universities and in industry.

3.4.1 *Framework for University-Industry Engagement*

Our framework for UI engagement drew from the personal experience of authors (who have more than four decades leading university-industry research programs) and consultations with eight highly experienced UI officers. We also drew ideas from the UI literature.

We identified key mission areas through which engagement occurs, and identified UI engagement activities that span mission areas. We then articulated the various UI engagement activities that fall within each area. After creating a framework for activities, we then developed a framework for UI foundations representing the core determinants of how an industry engagement strategy is executed.

Our activities and foundations framework, in draft form, stimulated conversations with engagement leaders at four U.S. and two non-U.S. universities as well as two companies. Their feedback was used to refine the framework, as well as to add activities that were initially overlooked. Interviews were also used to elaborate on characteristics of UI collaboration that were perceived as positive, promoting productive relationships.

3.4.2 Review of Websites

To understand how universities represent UI engagement to industry and the public at large, we reviewed the content of websites at 65 university members of the UIDP organization, representing UIDP's most research active university members. Universities were selected based on prominence, as reflected in US News and World Report's ranking of national universities (top 100), total federal research funding (top 100) and membership in the Association of American Universities. UIDP membership is predominantly American and therefore our sample is weighted toward the characteristics of American institutions, but does not exclude non-American UIDP members. Among the U.S. universities, federally funded research expenditures averaged \$409 million and industry funded research expenditures averaged \$44 million, both in academic year 2019. On average, the 61 U.S. universities in the sample ranked 50th on average in US News.

For each selected university, we identified the webpage that best represents UI relations at the level of the university as a whole (i.e., not any individual school or college) by examining stated purpose, scope of activity, and staffing. These offices were then reviewed to create a schema of common models for organizing central UI engagement offices. Each office was then categorized accordingly. We also assembled titles for the offices and calculated frequencies of descriptive words used in the titles. Last, by examining office descriptions, staffing, and other associated offices at the universities, we classified the offices as to their level of authority for each of a set of eight UI activities.

3.4.3 Surveys

Industry engagement strategies were assessed among UIDP members by an online Qualtrics survey. Two survey instruments were used, one targeting UIDP industry members and the second targeting UIDP university members. The university instrument was divided into two sections, first collecting information on the university's organization of UI engagement (e.g., scope, name, and existence of a central office; responsible offices for various types of contract negotiations; and distribution/experience of industry relations officers) and second assessing perceptions of importance of various UI strategies. Perceptions were also obtained from open-ended

questions on effective strategies. Respondents were finally asked to categorize their universities by size and type (public versus private).

The industry instrument was also divided into two sections, first collecting information on respondents' individual job responsibilities for UI engagement, and then asking for perceived importance of various UI strategies, utilizing questions from the university survey. Respondents were also asked to categorize their companies by industry focus, and they were provided the opportunity to answer an open-ended question on effective strategies for UI engagement.

Participants were invited to participate by emails sent from the UIDP chief executive in March of 2021. A reminder was sent two weeks later to those who did not click on the survey link, and a second reminder was sent one month later to those who did not click on the second survey link.

Because the university survey asked respondents to describe specific features of their universities, invitations (and two subsequent reminders) were only sent to the designated university representative at each university, a total of 124 people. Designated representatives have lead responsibilities for UI engagement at their universities and can therefore speak with authority as to university strategy. The industry survey was sent to the entire UIDP industry distribution list, a total of 600 recipients.

Responses to the two surveys were tabulated. Responses to questions on perceived importance were compared between the two survey populations, industry and university.

3.5 Options, Solutions, and Impact

In this section we present our framework for engagement and results from our website analysis and surveys of UI representatives.

3.5.1 *Framework for Engagement*

Table 3.1 presents the first part of our framework, the activities for engagement. At the top level, the mission spanning activities are represented by strategy, values, organization, concentration, and grand challenges. These are areas often led by the university president (or senior leadership), academic senate and board of trustees/regents, spanning academic units and functional areas. Grand challenges and concentrations are knowledge integration communities: they are interdisciplinary and mission-spanning in execution, as described by Acworth (2008). Strategy and values define how the university prioritizes industry engagement and how it intends to resolve the barriers of orientation; organization defines how the university intends to resolve the barriers of transactions and how it utilizes intermediaries to facilitate engagement.

Table 3.1 Activities for university engagement with industry

University Strategy and Vision				
Strategy Linking All Vision Areas				
Values Supporting Industry Engagement				
Organization Integrating Units and Applications				
Concentrations in Industry Sectors				
Grand Challenges Meeting Societal Needs				
Mission Activities				
Education	Research	Clinical	Auxiliaries & Administration	Economic/Community Development
• Alumni affinity groups • Capstone courses • Career planning • Competitions • Continuing/Exec Ed • Course content • Industry curricula • Industry-focused student organizations • Industry lecturers • Internships, co-ops • Online program management	• Advisory boards • Co-location • Conferences, workshops • Directed services • Facilities & data • Industry affiliation • Integrated centers • Joint employment • Joint pursuit • Open software	• Clinical innovation • Clinic, lab & hospital integration • Industry sponsored trials • Insurer-provider networks	• Access to conference facilities • Joint development of products • Strategic purchasing	• Inclusive community development • Joint development of Innovation park/incubator • Licensed technology • SBIR/STTR support • Small business support • Standards creation • Startup investment • Technical assistance

Individual activities are categorized by five mission areas: education, research, clinical, auxiliaries, and economic development. Auxiliaries are administrative activities not directly tied to academics, such as purchasing, but which nevertheless entail significant industry engagement. Economic development represents the outward facing activities through which the university supports companies and communities. The clinical mission is relevant to those universities that provide healthcare services to the public through clinics and hospitals. Clinical UI activities include clinical trials and clinical innovation programs implemented within the university operated health system, as well as various alliances with clinics, labs, and hospitals.

Table 3.1 includes several activities that are not typically addressed in the UI literature. Industry participation in courses as part-time lecturers or full-time professors of practice provide a knowledge flow from industry to university. Industry also contributes to the construction of professionally focused curricula and, in the reverse direction, universities offer both for-credit and extension courses to train industry employees in new techniques and skills. Research relationships may at times include joint pursuit of government funding, or joint ventures that span university and

industry. These are examples of how we documented UI activities on multiple dimensions.

In Table 3.2, the foundations enumerate the many considerations in the execution of a university strategy for industry engagement. The policies provide the “guard rails” for acceptable behavior, balancing the twin goals of engagement and independence (e.g., via conflict of interest, academic freedom, and intellectual property policies). They also define how participants in UI engagement are rewarded for their efforts (e.g., through academic promotion and tenure). Agreements define the many types of transactions that occur between a university and a company. For each, authority is given to negotiate and execute to a university office, as enumerated in key units. Last, stakeholders are listed as to types of positions held by participants in UI relationships. Stakeholders include those who conduct the work entailed in the activities (Table 3.1); those who negotiate/execute agreements (Table 3.2); intermediaries to facilitate or initiate relationships; and beneficiaries.

The last component of our framework, Table 3.3, offers characteristics identified as important in our interviews. These are categorized into attitudes, experiences/capabilities, services, and attributes. Example characteristics include cooperation, mutual benefits, flexibility, and clear communication.

We highlight one area: concierge service for industry engagement. From our experience and from the literature, universities have many points of industry interaction, executed through different types of centralized offices as well as localized academic units. UI engagement officers may also vary in responsibility, in some cases having UI engagement as their prime responsibilities; for others, UI engagement may be one of many responsibilities. From the perspective of industry partners, it can be difficult to navigate through the university, knowing who is responsible for what, the steps involved in reaching an agreement, and how to develop extensive relationships that link multiple missions and disciplines.

In summary, our framework illuminates questions that universities consider in the development of their strategy for industry engagement.

Table 3.2 Foundations for UI engagement

Foundations			
Policies	Agreements	Key Units	Stakeholders
Academic Freedom	Confidentiality	Academic Units	Alumni
Conflict of Interest	Corporate Gifts	Advancement	Boards
Intellectual Property	Data & Material Transfer	Career Planning	Business Partners
Privacy	Employment Contracts	General Counsel	Community
Promotion & Tenure	Joint Ventures	Health System	Faculty
	Partnering	Industry Relations	Staff
	Purchasing	Innovation Parks	Students
	Research Service	Purchasing	
	Sponsored Research	Sponsored Projects	
	Technology Licensing	Technology Licensing	

Table 3.3 Characteristics of university-industry engagement

Characteristics			
Attitudes	Experience and capabilities	Services	Attributes
• Cooperation across university units, staff and faculty • Desire for persistent relationship • Emphasis on mutually beneficial relationships • Flexibility to overcome barriers • Litigation as last resort • Recognition of benefits from university engagement • Service focus in industry relationships	• Corporate engagement staff with relevant industry aspettante • Entrepreneurial experience • Professors of practice	• Availability of innovation park for companies • Bundling IP across multiple investigators • C-level engagement with companies • Concierge service for industry engagement • Proactive reach out to catalyze industry relationships • Stewardship over lifecycles of awards by an industry relations officer • Support for social purpose aims, including diversity and grand challenges	• Alliance managers empowered • Business intelligence and technology forecasting guiding decisions • Clearly communicated standards and norms for industry engagement • Comprehensive industry facing office integrating research, education and economic development • Strategy and vision for industry engagement stated by president or provost • University concentrations aligned with industries or technologies • University polices that align with industry engagement

3.5.2 Options

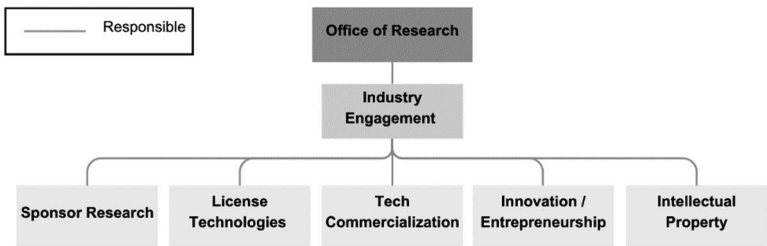
Though each university is somewhat unique in how it organizes UI engagement, we found five common approaches, as represented in Fig. 3.1 and described as follows:

Office of Research In this model, the relationship between corporations and the university is overseen by the office of research, and usually focuses on engagement through technology transfer, start-ups, corporate sponsored research, and sometimes economic development.

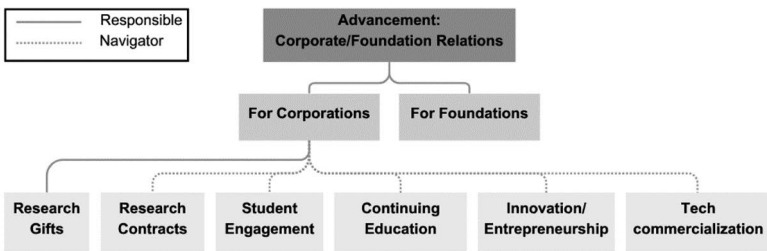
Economic Development/Government Relations This model is oriented toward governmental relationships that catalyze economic activity. It occurs in some public universities, specifically in locations where the local economy depends on government alliances with university support. In this model, the focus is on economic development through innovation centers, business parks, small-business support, and local/state government partnerships.

Corporate/Foundation Relations In this model, engagement is centered in the university advancement/development office. The office helps external organizations, both for-profit and not-for-profit, navigate the university, toward research, sponsorship, executive education, volunteering, or technology innovation. Such

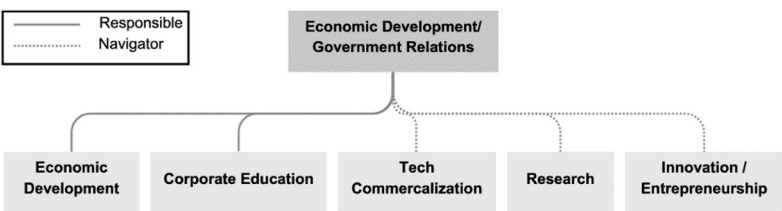
Office of Research: Tech Transfer



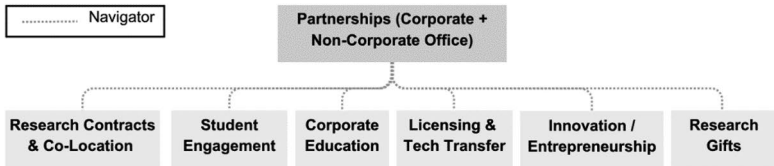
Economic Development/Government Relations



Corporate/Foundation Relations



Partnerships (Corporate + Non-Corporate)



Comprehensive (Spanning Missions)

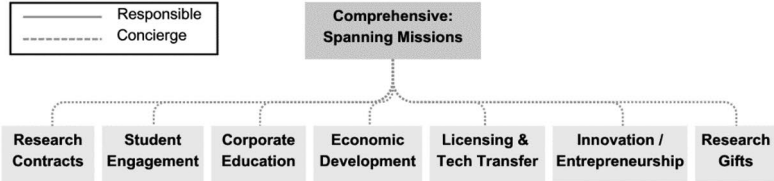


Fig. 3.1 Organizational models for central UI offices

offices can facilitate giving and donations, both financial and in-kind, as well as assist in contracted relationships.

Partnerships (Corporate + Non-Corporate) In this model, the office manages and improves strategic partnerships with external entities, including corporations, government laboratories, and independent research organizations. While they help corporations navigate in a very broad way through many different areas of the university, they usually focus on economic development, commercialization, and innovation.

Comprehensive (Spanning Missions) This model establishes a centralized corporate relations office, which acts as a concierge to guide corporations in their involvement and relationships with the university across many different areas. The office takes a holistic approach in engaging industry, and integrates different types of industry relationships through gifts and contracts, education, research, and economic development in a single office.

Other than the five common approaches, two universities showed no central office of any type. These were classified as decentralized.

Table 3.4 shows counts of how offices were classified by model. Based on our approach, we found that the partnership, comprehensive and office of research model were most common, each representing about ¼ of offices. Corporate and foundation relations offices were somewhat less common, representing about 20% of offices. The remaining offices were either classified as economic development or decentralized.

Looking deeper at how the offices function, Table 3.5 provides the classification according to authority for eight activities. Despite their central roles, offices were never fully comprehensive. Clinical and auxiliary activities were rarely mentioned, and offices rarely showed full authority for stated activities. A more common role was navigator or concierge. Whereas the offices clearly help create relationships in academic areas, they typically do not hold direct responsibility for execution, with responsibilities distributed among units that specialized in particular activities, such as contracting, clinical trials or gifts. While central offices offer general knowledge

Table 3.4 Classification by organizational model

Central Office Organization Model	#
Office of Research	16
Corporate/Foundation Relations	11
Economic Development/Government Relations	2
Partnerships (Corporate + Non Corporate Office)	17
Comprehensive (Spanning Missions)	17
Decentralized	2

of UI engagement, they tend to rely on cooperation with other units to turn relationships into formal agreements.

In their titles, 37% of offices used the word “relationships,” 28% “partnerships,” 26% “engagement,” and 14% innovation; 46% used the word “corporate” and 22% used the word industry. Titles convey the aims of offices. The word “corporate” defines a type of business entity, perhaps emphasizing larger and more established businesses than start-up companies (though those too may be incorporated). “Relationships” are broader and longer-lasting than “engagements.” “Partnerships” convey a stronger and deeper sense of mutuality than just relationships. In this manner, the name may convey a less-transactional and more strategic purpose, perhaps spanning a broad spectrum of interactions.

3.5.3 Solutions Reflected in Surveys

A total of 43 survey responses were received from the 124 university representatives invited to participate, for a 35% response rate. Of the 124 recipients, 49% opened the original invitation, 40% opened the first reminder and 39% opened the second reminder. The response rate was lower for industry representatives—43 responses out of 600 invitees—likely due to the breadth of the distribution list and lower likelihood of opening emails.

University respondents predominantly represented large universities (65% had more than 30,000 students and 69% more than \$500 million in annual research expenditures), the majority of which were public (69%). Industry respondents were employed within a highly diverse range of industries, the most common being chemical (15%) and software (13%). A total of 59% of industry respondents had postgraduation experience working for a university.

Responses demonstrated that UI engagement strategies are broad in scope but organizationally diffuse. As shown in Table 3.6, more than 80% of universities provide incubators, employ industry lecturers and professors of practice, offer space for

Table 3.6 Percent of universities utilizing UI engagement strategy (SE = standard error)

University Strategy	Percent	SE
Provide an incubator for start-up companies in non-life science technology?	89%	5%
Employ lecturers who are full-time employees at companies?	88%	6%
Employ full-time professors of practice, hired because of their industry experience?	84%	6%
Offer space for companies in university owned buildings?	84%	6%
Have a start-up investment program?	84%	6%
Provide an incubator for start-up companies in life sciences?	81%	7%
Operate or own an innovation or research park for industry?	68%	8%
Offer an industry affiliate program?	65%	10%
Employ tenured or tenure-track faculty who are jointly employed in industry?	63%	8%

companies in university buildings, and invest in start-ups. More than 60% have joint industry/university tenure-track faculty, offer an industry affiliate program and own or operate an innovation/research park for industry.

Organizationally, 84% of universities reported that they have an office designated to lead industry engagement, 39% of which report to the chief research office, 16% the advancement office, 13% to the president, 10% to the provost, and 23% to other (such as dual report to both the VP of research and advancement). Similar to our review of websites, stated office names emphasize “relationships,” “partnerships,” “engagement,” and “innovation” with applications to “corporate,” “industry,” “foundation” and “business,” and sometimes more broadly toward “external,” which might encompass industry and nonindustry partners (Table 3.7).

Respondents were asked to select which of 12 activities fell within the scope of the central office. Similar to our review of websites, respondents reported that responsibilities did not encompass all aspects of UI relationships—at most 9 of 12 areas were selected, and on average just 3½ areas were identified within the scope of responsibility. Most commonly, the scope included industry sponsored research (68%) and industry affiliations (65%), then secondarily technology licensing, start-up creation, industry sponsored clinical trials, co-op programs/internships, small-business technical support and industry related student organizations. More rarely, the offices also held responsibility for executive/continuing education, alumni affinity groups, career planning and for-credit courses targeting industry employees.

At a working level, Table 3.8 shows which university units had lead responsibility for negotiating terms and conditions for different types of agreements. As shown,

Table 3.7 Emphasis and application stated in central office names

Emphasis	Percentage		Application	Percentage	
	Survey	Website		Survey	Website
Relationships	23%	37%	Corporate	50%	46%
Partnerships	18%	28%	Industry	5%	22%
Engagement	36%	26%	Foundation	9%	20%
Innovation	27%	14%	Business	9%	8%
Technology	0%	6%	External	5%	2%
Economic	0%	5%			
Development	9%	3%			
Services	0%	3%			
Ventures	0%	3%			
Alliances	0%	2%			
Advancement	0%	2%			
Support	0%	2%			
Connect	0%	2%			
Exchange	0%	2%			
Outreach	0%	2%			
Portal	0%	2%			
Commercialization	5%	0%			

Table 3.8 Lead office for negotiating agreements by type (shown as % of responses)

Agreement Type	Sponsored Projects	Industry Relations	Technology Transfer	Individual Schools	General Counsel	Central Advancement	Clinical Trials Office	Other	Don't Know	Not Applicable
Industry Sponsored Research	50	12	12	3	12	3	0	9		
Research Services	29	11	8	18	13	0	0	13	5	3
Data & Material Transfer	27	6	39	6	6	0	0	12		
Membership Agreements	24	21	3	21	11	3	0	16	3	
Clinical Trials	24	11	8	11	3	0	29	3	3	11
Joint Ventures	18	26	5	5	16	5	0	8	8	8
Corporate Gifts	5	13	3	8	8	42	0	18	3	
Percentage Sum	177	100	78	72	69	53	29	79	22	22

Table 3.9 University relations officer (employed in industry): Average self-reported rating of activity importance in own job (5 points scale, SE = standard error)

Activity	Average importance	SE	Responses
Industry sponsored research	3.74	0.23	43
University affiliations	3.33	0.23	43
Technical assistance	3.11	0.19	36
Co-ops and internships	3.11	0.21	36
Technology licensing	3.08	0.23	36
Accessing specialized facilities or equipment	3.06	0.21	36
Career planning and student employment	2.81	0.22	36
Start-up creation	2.69	0.22	36
Executive and continuing education	2.28	0.21	36
Student organizations and competitions	2.17	0.18	36
Degree programs and for-credit courses targeted to industry employees	2.08	0.19	36
Alumni affinity groups for industry	1.97	0.18	36
Industry sponsored clinical trials	1.95	0.22	43

different universities assign lead responsibility to different units, depending on type of agreement, and there is no common standard for who holds the lead. The industry relations office takes the lead most often for joint venture agreements (highlighted in green), but in all other cases the lead office is most often something else, such as sponsored projects (industry sponsored research, research services and membership agreements), technology transfer, the clinical trials office or central advancement.

In addition to responsibilities divided among units, in a separate question respondents reported that industry relations officers were distributed among units. On average, 26% were estimated to be employed in schools/colleges, 25% in a central research office, 16% in central advancement, just 14% in a designated industry relations office outside of advancement or research, as well as 20% in other locations.

Just as industry relations officers in universities tend not to be all encompassing, university relations officers (employed in industry) also have limited job scope. Table 3.9 shows self-reported importance of various activities in their personal jobs. The highest rated activity on a 5-point scale was industry sponsored research, but even that only rated 3.74 on average. Nevertheless, 80% of respondents held responsibility for negotiating terms and conditions for sponsored research (Table 3.10).

Table 3.10 Percentage of university relations officers (employed in industry) having personal responsibility for negotiating terms and conditions

Contracting area	Average	SE	Responses
Industry sponsored research	80%	0.063	41
Membership agreements in industry alliance programs	71%	0.076	42
Data and material transfer	63%	0.079	41
Research services, not intended for publication	62%	0.079	42
Joint ventures with universities	61%	0 070	41
Technology licensing	54%	0 077	41
Corporate gifts	44%	0.077	41
Clinical Trials	24%	0 068	41

On the other hand, just 44% were responsible for negotiating corporate gift agreements and 24% were responsible for negotiating clinical trial agreements.

Our survey suggests that it would be quite rare for a university-relations officer to hold broad responsibility (as well as focus) for all areas of UI engagement. This might be explained by work responsibilities that extend beyond university relations as well as responsibilities that focus only on certain aspects of UI engagement.

3.5.4 *Impact*

Utilizing the frameworks developed in Sect. 3.5, industry and university respondents were asked to prioritize various aspects of UI engagement within the categories of attitudes, services, and attributes. For each category, respondents picked their three most important choices from a group of options. Both university and industry respondents most often selected “mutually beneficial relationships” and “flexibility to overcome barriers” as important attitudes, and selected litigation as last resort and service focus in industry relationships much less often. On the other hand, university respondents were more likely to select cooperation across university units and industry respondents more likely to select flexibility to overcome barriers (marked in yellow in the table).

Among services, both industry and university respondents most often selected “proactive outreach” and “concierge service.” On the other hand, university respondents were much more likely to select “C-level engagement” and industry respondents were more likely to select “bundling of intellectual property.” In this regard, university representatives appear to aspire toward higher level engagements whereas industry representatives appear to aspire toward projects that translate into technology transfer.

Last, with respect to attributes, both industry and university respondents most often selected alignment with industries, alignment in policies and provision of a comprehensive industry facing office. On the other hand, industry respondents were more likely to select empowered alliance officers and aligned university policies.

These results do not imply that other topics do not matter, but they do suggest guiding principles. Overall, almost all respondents agreed that pursuing mutually beneficial relationships was a top priority, as it was selected by more than 80% of respondents in each category. For industry respondents, flexibility to overcome barriers was a close second in priority (73.2%) and aligned policies was third (68.3%); for university respondents, concierge service (64.9%) and cooperation across units (63.9%) were second and third. While industry and university respondents differed in these secondary priorities, they are nevertheless aligned, as concierge and cooperation are means for achieving the industry aim of overcoming barriers.

Drilling down further, university respondents were asked for examples of effective industry engagement strategies at their universities as well as areas of improvement. Among these respondents, comments emphasized effective partnership and execution with both faculty and industry and effective communication to overcome bureaucratic hurdles and misunderstandings of academic culture. As one respondent stated, universities should: “Embrace industry engagement as part of core ethos. Efforts ranging from government relations to industry panels, workshops, councils, etc. are important to underscore ‘we’re open for business’. Operationally, translate industry or customer requirements into programs with the *authority* and *charge* for sustained delivery.”

Industry respondents were asked for effective engagement strategies used in universities and changes they would like to see in university services, policies or attitudes, in a single question. Comments frequently focused on intellectual property terms, talent pipeline and provision of industry relations officers that specialized in particular companies or industries, rather than academic discipline. As one industry respondent stated: “Flexible IP policies. University Partnership Managers who can make the process of understanding the universities research capabilities easier. University Partnership Managers specialized in my industry, to know what they need to know about working with us.”

3.6 Takeaways and Future Strategies

We have developed a framework for UI engagement that encompasses the many mission areas and touch-points through which industry and university work with each other toward mutual aims. A comprehensive plan, expressed through strategy, values, organization, concentrations and grand challenges, resting on the foundations of policies, agreements, units and stakeholders, can unify UI programs.

We found that universities pursue UI engagement in many different ways. Most have an office whose purpose is to centralize UI engagement, but in reality office scope is limited, largely acting as a concierge or navigator in limited areas. Clinical engagement and auxiliary engagement are rarely mentioned on websites for such central offices; UI agreements are negotiated in separate offices; and reporting structure and organization are widely variable. Given the range of models, it is no surprise that industry representatives can find universities difficult to engage.

As to priorities, pursuit of mutually beneficial relationships should be the focus. Flexibility to overcome barriers, policy alignment, concierge service, and cooperation across units are also important, in the eyes of industry and universities.

Our framework can help universities develop and implement strategies for UI engagement that are comprehensive and coherent. We recommend that universities express the various pieces of engagement (missions and touch points), how they are executed and organized, and the governing policies, at the level of the president and provost. Example policies would be those governing academic freedom, intellectual property, conflicts of interest, privacy and conflicts of interest, all of which bear on industry relationships. These should be stated and communicated as a unified whole, with detail to show how the university will work with industry toward their common aims, rather than as service relationships.

Mutual-respect should be a cornerstone of UI engagement, and universities should create the internal mechanisms to overcome obstacles to UI relationships, including relationship officers that fully understand the industries of their corporate partners. Relationship officers must also fully understand the internal workings and capabilities of university offices and academic programs. Universities need to create synergy through relationships with companies that span multiple missions and academic units while maintaining flexibility to adapt to the particular circumstances arising from disciplines, industry, and mission.

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